Beyond Mean-Variance: Goal-Based Portfolio Optimisation with Derivatives and Structured Products

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From a mental-accounting thesis to a scalable Monte-Carlo + CVaR engine

Companion technical paper to the interactive *Beyond Mean-Variance* optimiser

`https://sami-jeddou-behavioral-portfolio-optimizer.streamlit.app`

`https://github.com/SamiJeddou/behavioral-portfolio-optimizer`

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Abstract

This paper documents a research optimiser that builds goal-based portfolios beyond the classical mean-variance frame. It re-implements and extends a 2012 MSc thesis (USI Lugano) that replicated the mental-accounting models of Das & Statman (2009) and Das, Markowitz, Scheid & Statman (2010). Two engines are provided: an exact *grid* optimiser that reproduces the thesis results on a discretised state space, and a scalable *Monte-Carlo + CVaR* optimiser that escapes the curse of dimensionality through copula scenarios and a linear program. Both price derivatives and structured products arbitrage-consistently, and an out-of-sample back-test holds the chosen portfolio on later, unseen data. Beyond a faithful re-implementation, the paper records two substantive corrections relative to the thesis—the enforcement of the Expected-Shortfall constraint and the fair pricing of structured notes—and reports results, validation and limitations.

Keywords: behavioral portfolio theory, mental accounting, goal-based investing, portfolio optimisation, Value-at-Risk, Expected Shortfall, Conditional Value-at-Risk, derivatives, structured products, Monte-Carlo simulation, copula, out-of-sample backtesting.

JEL classification: G11, G41, G17, C61, C63.

1 Background and motivation

Markowitz mean-variance (MV) analysis treats risk as a single portfolio variance and assumes the investor optimises one aggregate trade-off. Behavioural evidence instead suggests that people manage wealth in *mental accounts* tied to distinct goals, each with its own attitude to loss. Das, Markowitz, Scheid and Statman (2010) formalise this and show a precise equivalence between the mental-accounting (MAT) specification and classical MV; Das and Statman (2009) extend the goal-based optimisation to derivatives and non-normal returns.

The author's 2012 MSc Finance thesis (USI Lugano, supervised by Prof. Enrico De Giorgi) replicated both methodologies in R and extended them with additional derivative simulations and parameter analysis. The present work re-implements that foundation in Python and extends it with live market data, a wider library of structured products, an out-of-sample back-test, and a scalable scenario-based engine.

2 The mental-accounting framework

An investor allocates wealth across goal sub-portfolios. Each goal carries a return threshold H and a maximum acceptable probability α of falling short of it. With weights w on the simplex $\Delta = \{w : w^\top \mathbf{1} = 1,\ w \geq 0\}$ and portfolio return $r_p = w^\top r$, the Value-at-Risk form of the goal problem is

$$\max_{w \in \Delta}\ w^\top \mu \qquad \text{s.t.} \qquad \mathbb{P}(r_p \leq H) \leq \alpha. \tag{1}$$

A tail-average (Expected-Shortfall) variant replaces the shortfall probability with a floor on the conditional mean of the loss tail,

$$\max_{w \in \Delta}\ w^\top \mu \qquad \text{s.t.} \qquad \mathbb{E}[r_p \mid r_p < H] \geq L. \tag{2}$$

The key theoretical result is the MVT–MAT equivalence: each goal's optimum is mean-variance efficient and can be written as $\max_w\ w^\top \mu - \frac{\lambda}{2}\, w^\top \Sigma w$ for an implied risk-aversion coefficient λ fixed by the binding shortfall constraint. The behavioural specification can therefore be read back as a classical MV problem with a goal-specific λ, and the optimiser reports an implied λ for every goal to reconcile the two views of the same portfolio.

3 The exact grid optimiser

Asset returns are modelled as multivariate normal with mean vector μ and covariance Σ. The engine enumerates candidate portfolios on a discretised grid: each asset's return is sampled over $\mu \pm k\sigma$ in m steps, and weights are searched on a simplex grid of resolution m'. For n assets the state space grows as roughly m^n—the result is exact and thesis-faithful, but practical only to about four assets.

Two Expected-Shortfall modes are offered. The *thesis-faithful* mode reproduces the original R behaviour, in which the refinement step enforces the VaR penalty and the ES condition is applied at grid-eligibility; the *rigorous-ES* mode instead maximises return strictly subject to (2) and recovers expected return that the thesis method leaves on the table (§9). Four resolutions are available—Fast, Standard and High, plus a custom *Turbo* coarse-to-fine solver that matches High-precision VaR to within a few basis points at about sixty times the speed. For five or more assets a differential-evolution search replaces the exhaustive grid.

4 Derivatives and structured products

Sixteen instruments are priced arbitrage-consistently with Black–Scholes—vanilla options, spreads, collars, capital-guaranteed and barrier notes, and certificates—alongside a builder for configuring bespoke payoffs. The Black–Scholes call and its standard inputs are

$$C(S_0, K, T) = S_0\, \Phi(d_1) - K e^{-rT}\, \Phi(d_2), \qquad d_1 = \frac{\ln(S_0/K) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \tag{3}$$

Under each modelled outcome the instrument's payoff is evaluated and folded into the portfolio return, so asymmetric, non-normal payoffs enter the optimisation directly rather than through a variance proxy—precisely the behaviour the classical mean-variance frame cannot represent. A structured note is valued as the present value of its bond floor plus its embedded option(s), not at par (§9).

5 The scalable Monte-Carlo + CVaR engine

To escape the m^n state space, the second engine samples S joint return scenarios through a copula—Gaussian, or Student-t for tail co-movement (assets crashing together)—prices its supported derivatives under each scenario, and solves the goal as a linear program. Tail risk is measured coherently with Conditional Value-at-Risk (Expected Shortfall). The Rockafellar–Uryasev representation is

$$\mathrm{CVaR}_\alpha(L) = \min_{\zeta\in\mathbb{R}} \Big\{ \zeta + \tfrac{1}{\alpha}\,\mathbb{E}\big[(L-\zeta)_+\big] \Big\}, \tag{4}$$

which is linear in scenario form. Maximising expected return subject to a CVaR floor then becomes a single linear program in the weights, the auxiliary level ζ and the per-scenario slacks; the complete formulation, with derivative columns and its relationship to the grid, is derived in the accompanying addendum.[1] Cost is linear in the number of assets and effectively free in the number of scenarios, so the engine scales to dozens of holdings with several overlays at once. It is solved with the HiGHS LP solver, and a single set of common random numbers is reused across floors to give a smooth, comparable efficient frontier.

A note on comparability. The grid uses a fixed-threshold Expected Shortfall, $\mathbb{E}[r_p \mid r_p < H]$, whereas the linear program uses α-CVaR. These coincide only when H equals the α-quantile; otherwise they are two valid but different tail measures. The scalable engine therefore *complements* the exact grid reference—it does not replace it.

Instrument coverage. The full sixteen-instrument library—including the structured notes and the path-dependent barrier note—is available in the grid engine. The scalable engine currently covers the family expressible as single-underlying options with up to two strikes: vanilla calls and puts, straddles and strangles, and bull/bear vertical spreads. The remaining payoffs are either multi-leg or carry floor/cap/premium parameters, and the barrier note is path-dependent and so cannot be resolved from a single horizon scenario; extending the engine to the static multi-leg payoffs is straightforward, the path-dependent ones less so.

6 Out-of-sample back-testing

Does the optimisation hold up on unseen data? The back-test builds optimal weights on a construction window, then holds those fixed weights (buy-and-hold) through a later evaluation window, marking any derivative to market at each step with a shrinking maturity. It compares what the model expected with what actually happened—realised versus expected return and volatility, and the loss-threshold outcome—and reports the realised alpha, beta and R^2 of each holding and of the portfolio against a chosen benchmark, with an optional CAPM (ex-ante) alpha. Both a no-derivative portfolio and a with-derivative portfolio are tracked side by side.

7 Results

Table 1 reports the effect of a single derivative overlay at a fixed downside limit on the three-asset thesis data set (annual returns 5/10/25%, volatilities 5/20/50%, correlation $\rho = 0.4$). Under the grid (VaR) engine, adding an at-the-money straddle lifts the optimal expected return from 9.76% to 11.29% (+1.53 **pp**) at the same −10%/5% limit, lifting skewness from 0 to +0.99; the gain is paid in variance (vol 19.56%) and collected in positive skewness—the dimension mean-variance cannot see. Constrained back to the no-derivative variance, the same overlay earns only 8.95% (−0.81 pp): the edge is the skewness it buys, not free return. Under the scalable (α-CVaR) engine, a protective put (struck 0.8×, one year) lifts the optimum from 11.00% to 15.73% (+4.73 **pp**) at a −20% CVaR floor, raising skewness from near zero to +0.71.

[1] S. Ben Jeddou, *A Scalable Monte-Carlo / CVaR Linear-Program Optimiser for Mental-Account Portfolios with Many Securities and Derivatives* — companion addendum to this paper. Available at `https://github.com/SamiJeddou/behavioral-portfolio-optimizer/blob/main/addendum.pdf`.

Engine / constraint	Overlay	$\mathbb{E}[r]$	$\Delta\mathbb{E}[r]$	skew
Grid, VaR (H=−10%, α=5%)	none	9.76%	—	0
Grid, VaR (H=−10%, α=5%)	ATM straddle	11.29%	+1.53 pp	+0.99
Scalable, α-CVaR (L=−20%)	none	11.00%	—	≈ 0
Scalable, α-CVaR (L=−20%)	protective put	15.73%	+4.73 pp	+0.71

Table 1: Effect of a derivative overlay at a fixed downside limit (three-asset thesis data). The VaR and CVaR cases answer different questions and are not directly comparable; each constraint binds at its optimum.

Out of sample, the two engines tell a consistent story. On a 2017 evaluation window the plain portfolio realised 28.19% and the derivative-enhanced portfolio 47.59% (∼19 pp more), with realised CAPM alpha against the S&P 500 of +1.99% versus +7.95%. Read honestly, realised returns ran above the model's expected ∼23–25% because 2017 was a strong trending year for these names; the point is the relative behaviour—the overlay added both market exposure and genuine non-linear return while holding the downside intact.

8 Validation

Every figure is checked rather than asserted.

- The Monte-Carlo optimum lies on the exact-grid mean-variance frontier to within a few basis points.
- Under the Gaussian copula the scenario Expected Shortfall matches the closed-form Normal value
$$\mathrm{ES}_\alpha = \mu_p - \sigma_p \frac{\phi\big(\Phi^{-1}(\alpha)\big)}{\alpha},$$
for example −28.4% versus −28.6%, confirming the generator and the tail estimate.
- The frontier is monotone, the ES floor binds at the optimum, and infeasible floors are flagged rather than silently relaxed.
- All thesis result tables reproduce within discretisation noise, end to end (R to Python).

Modelling horizon. Weights are optimised for a one-year hold (μ and σ are annualised), which lines up with an annual mandate, risk budget and yearly rebalancing; the back-test then bridges this single-period model horizon to a real, user-chosen holding window.

9 Changes and corrections relative to the 2012 thesis

The implementation is a faithful re-creation of the thesis; the differences are recorded here for traceability.

Corrections.

(a) *Expected-Shortfall enforcement.* In the original R code the gradient-refinement step maximises return under a Value-at-Risk penalty, and the Expected-Shortfall condition enters only at grid-eligibility—so the reported "ES-constrained" optima are not strictly ES-feasible after refinement. The app reproduces this thesis-faithful behaviour by default for comparability, and adds a *rigorous-ES* mode that maximises return strictly subject to (2). The rigorous mode stays ES-feasible throughout and recovers expected return the thesis method leaves unused. Validated against an independent brute-force ES maximum, a put overlay at $L = -15\%$ attains 15.5% under the rigorous mode versus 13.2% under the thesis-faithful method (+2.3 pp), and a straddle overlay at $L = -20\%$ attains 20.3% versus 11.9% (+8.4 pp); the thesis-faithful refinement can also finish ES-infeasible (for example reporting an Expected Shortfall of −25% against a −20% floor).

(b) *Fair pricing of structured notes.* An earlier headline figure for a capital-guaranteed note overstated its benefit because the note was valued at par rather than at fair, arbitrage-consistent value. Pricing every instrument with Black–Scholes—so a guaranteed note is the present value of its bond floor plus its embedded option—shrinks the apparent uplift dramatically—a par-priced headline of roughly +23% collapses to about +0.7 to +0.8 pp once the note is fairly priced—and shows the fairly-priced note is mean-variance-dominated yet can still improve the downside-constrained objective. That is a genuine behavioural effect rather than a pricing artefact, and it sharpens the Das–Statman case against the standard objection that fairly-priced structured products cannot help a rational investor.

Extensions. Live market data; an expanded library of sixteen instruments plus a custom-payoff builder; an out-of-sample back-test; the scalable Monte-Carlo + CVaR engine (full derivation in the addendum); the coarse-to-fine "Turbo" solver; and an AI-assisted narrative and glossary layer.

10 Limitations and assumptions

The grid engine assumes multivariate-normal marginals and is exact only on small universes ($n \leq 4$); the scalable engine is approximate, with Monte-Carlo error of order $S^{-1/2}$ that is heavier in the tail than in the body and is mitigated by sample size, antithetic variates and common random numbers. Results depend on *scenario quality*: the copula choice and the estimation of μ, σ, Σ drive the answer, and estimation error worsens with n, so shrinkage or robust estimators belong in any production pipeline. Derivatives are priced under Black–Scholes assumptions (constant volatility, frictionless hedging). The model is single-period; the back-test bridges to a real holding window but does not re-optimise within it. The scalable engine also exposes a subset of the instrument library—vanilla options and vertical spreads—with structured notes and the path-dependent barrier remaining grid-only (§5). Finally, the VaR/CVaR surrogacy of §5 means the two engines answer related but distinct questions. None of these is a barrier to the construction; each is a calibration responsibility.

11 Implementation

The optimiser is written in Python (NumPy, SciPy, the HiGHS linear-program solver and copula sampling) behind a Streamlit interface with Plotly and Altair charts. An AI-assisted layer turns each result into a plain-language narrative and powers an interactive glossary of risk measures and portfolio theory. The numerical pipeline is open and reproducible: copula scenarios → Black–Scholes pricing → CVaR linear program → efficient frontier → interactive app and PDF report.

Research and educational project. Not investment advice. The interactive application reproduces all figures referenced above and exports per-run PDF reports.

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